

Partitioning 3-homogeneous latin bitrades: a Lean blueprint

Carlo Härmäläinen

September 11, 2026

Contents

1	Introduction	2
2	Latin bitrades	3
3	The Euclidean triangle group	5
4	The three-colouring	6
5	Transversals	7
6	The worked example	8

Chapter 1

Introduction

This blueprint links the statements of *Partitioning 3-homogeneous latin bitrades* (arXiv:0710.0938) to their formal counterparts in the Lean 4 project `Partitioning3Homogeneous`. Every numbered statement below names the Lean declaration that formalises it, and the dependency graph records which statements are used to prove which.

Two conventions differ between the paper and Lean.

- The paper lets permutations act on the right, so $x\tau_1\tau_2$ means “apply τ_1 , then τ_2 ”. Lean’s `Equiv.Perm` composes right to left, so the same composite is written `tau2 * tau1`. In particular (T1), $\tau_1\tau_2\tau_3 = 1$, is the Lean statement `tauSymbol * tauColumn * tauRow = 1`.
- The paper works with $n \times n$ arrays. Lean works with arbitrary finite index types for rows, columns, and symbols, and 3-homogeneity is required of every element of those types. A bitrade with empty rows must be re-indexed before the main theorem applies.

Chapter 2

Latin bitrades

Definition 2.1 (Partial Latin square). A *partial Latin square* is a set $T \subseteq A_1 \times A_2 \times A_3$ of entries (i, j, k) , read as “symbol k in row i and column j ”, such that two entries agreeing in any two coordinates agree in the third.

Definition 2.2 (Latin bitrade; Definition 2.1). Let $T^\circ, T^\otimes$ be partial Latin squares. The pair $(T^\circ, T^\otimes)$ is a *bitrade* if

- (R1) $T^\circ \cap T^\otimes = \emptyset$;
- (R2) for all $(a_1, a_2, a_3) \in T^\circ$ and all $r \neq s$ there is a unique $(b_1, b_2, b_3) \in T^\otimes$ with $a_r = b_r$ and $a_s = b_s$;
- (R3) the same with T° and $T^\otimes$ exchanged.

In Lean, (R2) and (R3) are `HasUniqueMates positive negative` and `HasUniqueMates negative positive`.

Definition 2.3 (k -homogeneous). A bitrade is *k -homogeneous* if every row and every column of T° contains k entries and every symbol occurs k times in T° .

Definition 2.4 (Transversal). A set $X \subseteq T^\circ$ is a *transversal* if it meets each row and each column of T° in exactly one entry and $|\{k \mid (i, j, k) \in X\}| = |X|$. `IsThreeTransversalPartition` packages three transversals which are pairwise disjoint and cover T° .

Definition 2.5 (The mate bijections β_r). For $r \in \{1, 2, 3\}$ define $\beta_r : T^\otimes \rightarrow T^\circ$ by $(a_1, a_2, a_3)\beta_r = (b_1, b_2, b_3)$ if and only if $a_r \neq b_r$ and $a_i = b_i$ for $i \neq r$. By (R2) and (R3) each β_r is a bijection. In Lean the bijections are built from the unique-mate clauses with `mateEquivBy`; the inequality $a_r \neq b_r$ follows from (R1).

Definition 2.6 (The τ_i representation; Equation (1)). Define permutations of T° by

$$\tau_1 = \beta_2^{-1}\beta_3, \quad \tau_2 = \beta_3^{-1}\beta_1, \quad \tau_3 = \beta_1^{-1}\beta_2,$$

maps applied from left to right. The cycles of τ_1, τ_2, τ_3 are the rows, columns, and symbols of T° .

Lemma 2.7 ((T1) and coordinate preservation). $\tau_1\tau_2\tau_3 = 1$, and τ_1 preserves rows, τ_2 preserves columns, τ_3 preserves symbols.

Proof. Unfold the definitions; consecutive β 's cancel. Each τ_i is a composite of two maps that both fix the i th coordinate. $\square$

Lemma 2.8 ((T3): fixed-point-freeness). *Each τ_i is fixed-point-free.*

Proof. If $x\tau_1 = x$ then $y = x\beta_2^{-1} \in T^\otimes$ satisfies $y\beta_2 = y\beta_3 = x$, so y agrees with x in all three coordinates and $x \in T^\circ \cap T^\otimes$, contradicting (R1). $\square$

Lemma 2.9 (Order three; Equation (4)). *If $(T^\circ, T^\otimes)$ is 3-homogeneous then $\tau_i^3 = 1$ for $i = 1, 2, 3$.*

Proof. τ_1 permutes the three entries of each row without fixed points, so it acts on each row as a 3-cycle. The Lean proof first rules out a 2-cycle inside a 3-element fibre, then shows $x, x\tau_1, x\tau_1^2$ exhaust the fibre. $\square$

Theorem 2.10 (Drápal; Theorem 2.4). *A bitrade is equivalent up to isotopism to three permutations satisfying (T1), (T2), (T3), and it is primary if (T4) also holds.* Not formalised: the main theorem does not need it. Condition (T2) is checked only for the worked example.

Chapter 3

The Euclidean triangle group

Definition 3.1 (The group Γ ; Equation (3)). The paper takes $\Gamma = \langle \rho_1, \rho_2, \rho_3 \mid \rho_1^3 = \rho_2^3 = \rho_3^3 = \rho_1\rho_2\rho_3 = 1 \rangle$, the orientation-preserving $(3, 3, 3)$ triangle group acting on the triangular tessellation of the plane. Lean uses the concrete model $\mathbb{Z}[\omega] \rtimes \mathbb{Z}/3$, with $\mathbb{Z}[\omega] \cong \mathbb{Z}^2$ in the basis $(1, \omega)$ and $\mathbb{Z}/3$ acting by multiplication by ω . The generators are the affine maps

$$\rho_1: z \mapsto \omega z, \quad \rho_2: z \mapsto \omega z + 1, \quad \rho_3: z \mapsto \omega z + 1 + \omega,$$

anticlockwise rotations by $2\pi/3$ about the three vertices $0, (2 + \omega)/3, (1 + 2\omega)/3$ of one triangle.

Lemma 3.2 (Relations). $\rho_1^3 = \rho_2^3 = \rho_3^3 = \rho_1\rho_2\rho_3 = 1$ in the concrete model.

Proof. By computation (`decide`). □

Lemma 3.3 (Rotations are conjugate to generators). *Every element of Γ whose rotational component is ω is conjugate by a translation to one of ρ_1, ρ_2, ρ_3 . This is the algebraic replacement for Cases 1–4 in the proof of Lemma 4.3 of the paper.*

Proof. Conjugating $z \mapsto \omega z + (m + n\omega)$ by the translation by $p + q\omega$ gives $z \mapsto \omega z + (m + p + q) + (n + 2q - p)\omega$. The residue $m + n \bmod 3$ is invariant and equals $0, 1, 2$ on ρ_1, ρ_2, ρ_3 ; choose p, q to reach the representative. □

Proposition 3.4 (The representation θ ; Equation (4)). *Let a, b, c be elements of a group G with $a^3 = b^3 = c^3 = abc = 1$. There is a homomorphism $\theta: \Gamma \rightarrow G$ with $\rho_1 \mapsto a, \rho_2 \mapsto b, \rho_3 \mapsto c$.*

Proof. Put $u = ba^{-1}$ and $v = aua^{-1}$. From $b^3 = 1$ one gets $ava^{-1} = (uv)^{-1}$, and from $c^3 = 1$ that u and v commute. So (u, v) is a lattice representation on which conjugation by a acts as multiplication by ω , and the universal property of the semidirect product gives θ . □

Chapter 4

The three-colouring

Lemma 4.1 (Equivariant orbit maps; Lemma 4.1). *Let $\theta: \Gamma \rightarrow G$ be a homomorphism, X a transitive Γ -set with base point x_0 , and Y a G -set with base point y_0 . If θ maps the stabiliser of x_0 into the stabiliser of y_0 , then $\gamma \cdot x_0 \mapsto \theta(\gamma) \cdot y_0$ is a well-defined equivariant map $X \rightarrow Y$. The stabiliser hypothesis is exactly what the paper's proof of Lemma 4.1 leaves implicit; there it holds because Γ acts freely on the shaded triangles.*

Proof. If $\gamma_1 x_0 = \gamma_2 x_0$ then $\gamma_2^{-1} \gamma_1$ fixes x_0 , so $\theta(\gamma_2^{-1} \gamma_1)$ fixes y_0 and the two candidate values agree. Equivariance is the same computation. $\square$

Lemma 4.2 (Stabilisers are translations; Lemma 4.3). *Let Γ act on a set D so that ρ_1, ρ_2, ρ_3 have no fixed points. Then every point stabiliser is contained in the translation subgroup $\mathbb{Z}[\omega]$.*

Proof. An element with rotational component ω fixing x is conjugate by a translation t to some ρ_i , which then fixes $t \cdot x$. The component ω^2 case follows by taking inverses. $\square$

Theorem 4.3 (Three-colouring; Definition 4.2 and Corollary 4.5). *Let a, b, c be fixed-point-free permutations of a set D with $a^3 = b^3 = c^3 = abc = 1$. Then there is a colouring $\chi: D \rightarrow \mathbb{Z}/3$ with $\chi(ax) = \chi(bx) = \chi(cx) = \chi(x) + 1$ for all x . Neither (T2) nor transitivity (T4) is needed: the colouring is built on each orbit separately.*

Proof. Let Γ act on D through θ . On the orbit of x_0 define $\chi(\gamma \cdot x_0)$ to be the rotational component of γ . By Lemma 4.2 the stabiliser of x_0 has trivial rotational component, so Lemma 4.1 applied to the projection $\Gamma \rightarrow \mathbb{Z}/3$ shows χ is well defined. Each ρ_i has rotational component 1, so it advances χ by one. $\square$

Chapter 5

Transversals

Lemma 5.1 (Colour classes are transversals; Lemma 4.6). *Let τ_1, τ_2, τ_3 be fixed-point-free permutations of order three of a finite set D with $\tau_3\tau_2\tau_1 = 1$ (Lean order), preserving coordinates row, col, sym whose fibres all have three elements. Then the three colour classes of Theorem 4.3 each meet every fibre of every coordinate exactly once, are pairwise disjoint, and cover D .*

Proof. A fibre of row is $\{x, x\tau_1, x\tau_1^2\}$, whose colours are $\chi(x), \chi(x) + 1, \chi(x) + 2$. So each colour occurs exactly once per fibre. The same holds for columns and symbols. $\square$

Lemma 5.2 (Transfer to entries). *A partition of the entries of T° , regarded as an abstract dart set, into three coordinate transversals is a partition of T° into three transversals in the sense of Definition 2.4.*

Proof. Forget the membership proofs. The symbol-count clause follows from injectivity of the symbol map on each class. $\square$

Theorem 5.3 (Cavenagh; Theorem 1.1). *Let $(T^\circ, T^\otimes)$ be a 3-homogeneous bitrade. Then T° can be partitioned into three transversals.*

Proof. Apply Lemma 5.2 to τ_1, τ_2, τ_3 of Definition 2.6. The hypotheses are supplied by Lemmas 2.7, 2.8, and 2.9. $\square$

Lemma 5.4 (Tessellation on the torus; Lemma 3.5). *A 3-homogeneous bitrade defines a tessellation of the Euclidean plane by shaded and unshaded triangles, whose quotient is a torus. Not formalised: the geometric picture is replaced by Definition 3.1 and Lemma 4.2.*

Chapter 6

The worked example

Proposition 6.1 (Example 2.6 and its transversals). *The two arrays of Equation (2) form a 3-homogeneous bitrade, and the three sets displayed after Lemma 3.5 partition T° into transversals.*

Proof. Kernel computation on the twelve entries (**decide**). □

Proposition 6.2 (The printed τ_i satisfy (T1)–(T4)). *The permutations printed after Equation (2) are those of Equation (1), and they satisfy (T1), (T2), (T3), and (T4).*

Proof. Kernel computation. Transitivity is certified by an explicit word in the generators for every ordered pair of darts. □